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SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

IoT Exercise

PART 2 – 802.15.4

TESI DI LAUREA MAGISTRALE IN
TELECOMMUNICATION ENGINEERING

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1 | Exercise 2

1.1. Question 1

The goal is to compute the Probability Mass Function (PMF) of the output rate based on how many people are detected in each image frame, modeled by a Poisson distribution. The Poisson distribution models the probability of a given number of events occurring in a fixed interval.

The PMF is defined as:

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

where:

- $\lambda=0.15$: average number of people per frame
- $k=0,1,2,\dots$: observed number of people

Plug in $\lambda = 0.15$:

- $P(X=0)$:

$$P_0 = \frac{(0.15)^0}{0!} e^{-0.15} = 1 \cdot e^{-0.15} \approx 0.8607$$

- $P(X=1)$:

$$P_1 = \frac{(0.15)^1}{1!} e^{-0.15} = 0.15 \cdot e^{-0.15} \approx 0.1291$$

- $P(X \geq 2)$:

$$P_2 = 1 - P_0 - P_1 = 1 - 0.8607 - 0.1291 = 0.0102$$

People Count	Probability	Output Size
0	$P(r = r_0) = P(X = 0) = 86.07\%$	1 kB
1	$P(r = r_1) = P(X = 1) = 12.91\%$	3 kB
≥ 2	$P(r = r_2) = P(X \geq 0) = 1.02\%$	6 kB

Table 1.1: Categorize People Count

1.2. Question 2

Slot Duration:

$$T_s = \frac{L}{R} = \frac{128 \times 8 \text{ bits}}{250 \times 10^3 \text{ bps}} = \frac{1024}{250000} = 4.096 \text{ ms}$$

The minimum required rate is $r_{\min} = \frac{1KB}{10s} = \frac{8000bits}{10s} = 800[bit/s](\text{person_count} = 0)$.

Thus, the beacon interval duration can be found by imposing:

$$BI = \frac{L}{r_{\min}} = \frac{128Bytes}{800bit/s} = 1.28s$$

There are 3 camera nodes, each allocated $\frac{6KB}{1KB} = 6$ slots:

$$N_{CFP} = 3 \times 6 = 18 \text{ slots}$$

There are 18 slots in total, each lasting 4.096 ms:

$$T_{\text{active}} = (18 + 1) \times 4.096 \text{ ms} = 77.824 \text{ ms}$$

So the time of inactive is:

$$T_{\text{inactive}} = BI - T_{\text{active}} = 1.28 \text{ s} - 77.824 \text{ ms} = 1.202176 \text{ s} = 1202.176 \text{ ms}$$

$$\text{Duty cycle} = \frac{T_{\text{active}}}{BI} = \frac{77.824 \text{ ms}}{1.28 \text{ s}} = 6.08\%$$

The system is active for only about 6.08% of the time.

1.3. Question 3

To keep the system duty cycle below 10%, we require:

$$\frac{T_{\text{active}}}{BI} \leq 10\%$$

So,

$$\frac{(N_{CFP} + 1) * T_s}{BI} \leq 10\% \Rightarrow \frac{(6slot \cdot N_{camera} + 1) * T_s}{BI} \leq 10\%$$

We got:

$$6 \cdot N + 1 \leq \frac{0.1 \cdot 1.28 \text{ s}}{4.096 \text{ ms}} = 31.25 \rightarrow N \leq 5.042 \rightarrow \mathbf{N=5}$$

The system currently has 3 cameras, so we can add: **2** additional cameras.

